

EKKATI LAXMAN REDDY

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SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn), SQL, Java
Statistics & ML: Hypothesis Testing, Regression (Linear, Logistic), Decision Trees, K-Means, Hierarchical Clustering, Classification, EDA, Feature Engineering, PCA, Model Evaluation, Predictive Modeling
Generative AI & RAG: LLMs, RAG Pipeline Design, Semantic Search, Vector Search, ChromaDB, LangChain, Groq API, Prompt Engineering (Zero-shot, Few-shot, Chain-of-Thought), Transformers
Data Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI (concepts), Streamlit, Excel
Tools & Platforms: HuggingFace, Selenium, TestNG, Docker, Jenkins, Git/GitHub, AWS/GCP/Azure (concepts)
Soft Skills: Problem-Solving, Communication, Teamwork, Attention to Detail, Learning Agility, Adaptability

PROJECTS

Multi-Phase AI Retrieval & Analytics System | Live ↗ Feb – Mar 2026

Python · NLP · LLMs · RAG · Semantic Search · ChromaDB · Groq · BM25 · NetworkX · Streamlit

- Architected a 3-phase RAG pipeline (Dense → Hybrid → Graph RAG) over 500 arXiv papers, combining BM25 sparse + bi-encoder dense retrieval fused via RRF and cross-encoder reranking — achieving **Recall@5 of 0.76** and **43% MRR improvement** over single-retriever baseline across 30 evaluation benchmarks.
- Developed **semantic search** over a large document corpus using vector embeddings and ChromaDB — designing a Graph RAG layer with NetworkX BFS-expanding 1-hop citation neighbours for deeper cross-paper reasoning and context-aware retrieval.
- Engineered a 3-signal confidence scorer (retrieval sigmoid + LLM self-rating + chunk agreement) and a **9-chart EDA dashboard** with PCA embedding projection — documenting analytical findings and making the pipeline fully interpretable end-to-end.

Customer Segmentation for Retail Store Analysis Jun 2024

Python · Pandas · NumPy · Scikit-learn · K-Means · Hierarchical Clustering · Matplotlib · Seaborn

- Executed **data collection, cleaning, and preprocessing** on retail customer data — addressing 100% missing values and normalizing features to enhance clustering reliability, maintaining data quality and integrity throughout the pipeline.
- Applied **K-Means and Hierarchical clustering** algorithms to segment customers into distinct groups based on spending behaviour and income — conducting EDA to identify trends and patterns and translating findings into actionable business insights.
- Enhanced customer insights through **data visualization** using Matplotlib and Seaborn — improving interpretability of analytical outputs by **40%** and preparing clear reports to communicate findings effectively to business stakeholders.

Traffic Analysis Dashboard Jan – Feb 2025

Tableau · Python · Google Sheets API · Geoapify API

- Designed an automated real-time traffic dashboard in Tableau integrating Google Sheets and Geoapify API data streams — visualizing route efficiency, commercial hotspots and key traffic metrics, improving navigation insights by **80%**.
- Built an end-to-end **data extraction and transformation pipeline** across multiple APIs, automating dataset refresh cycles and reducing manual processing effort by **80%** — maintaining clear documentation of data processes and analytical findings.
- Enabled geospatial decision-making by mapping traffic patterns and location intelligence, contributing to a **15% increase in operational efficiency**.

INTERNSHIP

Programmer Analyst Trainee Jan 2026 – Present

Cognizant Technology Solutions (GenC Program)

- Built and shipped an end-to-end test automation backend — designed a fault-tolerant Selenium TestNG framework with structured exception handling, WebDriverWait synchronization, and Assert-based validation covering edge cases beyond the happy path.
- Integrated external REST APIs using REST Assured — writing and validating GET, POST, PUT, PATCH, DELETE requests with JSON/XML schema validation, status code assertions and data-driven parameterization via CSV and JSON.
- Worked across a full backend-aligned stack: Java, Maven, Git/GitHub, Jenkins CI/CD, Docker, BDD Cucumber and Agile sprint workflows — collaborating with senior engineers on system design and iterative delivery under real deadlines.

EDUCATION

Lovely Professional University Phagwara, Punjab

Bachelor of Technology, Computer Science and Engineering | CGPA: 8.1

Aug 2022 – Present

Sri Chaitanya Junior College Hyderabad, Telangana

Intermediate | 96.8%

Apr 2021 – May 2022

CERTIFICATIONS & ACHIEVEMENTS

Associate Data Practitioner Certification | Credly Apr 2026

Associate Data Practitioner Certification | Credly Mar 2026

Data Visualization with Tableau | Coursera Nov 2024

Achieved 5-Star rating in SQL | HackerRank Feb 2025